Summary in Graph

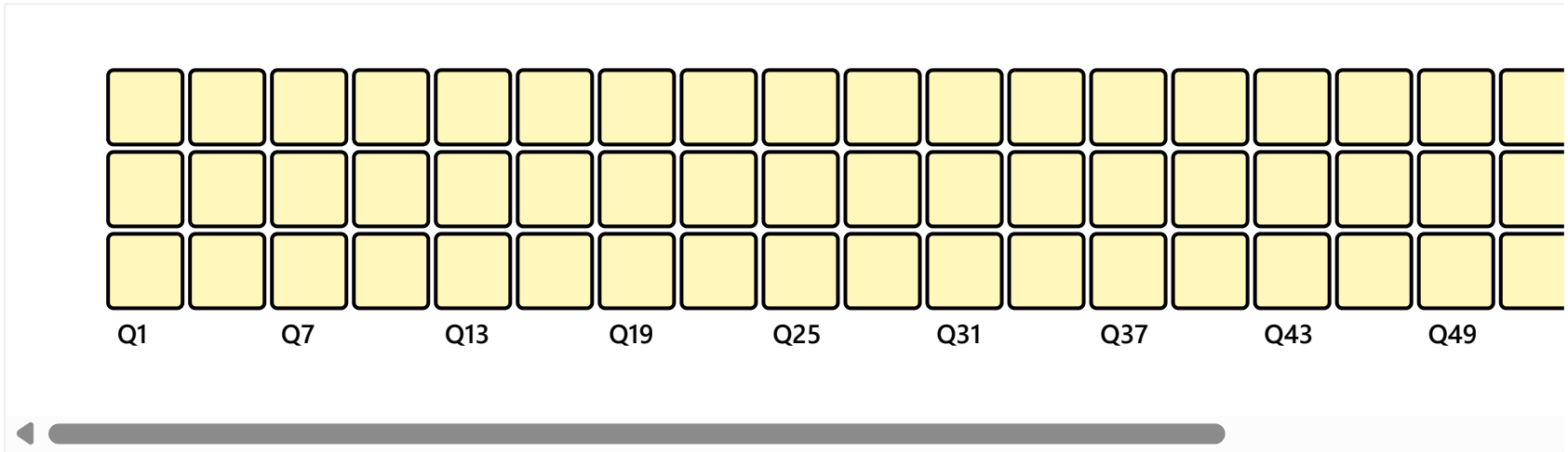
Exam Summary (GO Classes CS Test Series 2026 | MOCK GATE | Test 12)

Qs. Attempted:	0 0 + 0	Correct Marks:	0 0 + 0
Correct Attempts:	0 0 + 0	Penalty Marks:	0 0 + 0
Incorrect Attempts:	0 0 + 0	Resultant Marks:	0 0 + 0

Total Questions:	65 30 + 35
Total Marks:	100 30 + 70
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (76) out of 755 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

The sequence S is defined by $S_n = 2\left(S_{n-1}\right) - 4$ for each integer $n \geq 2$. If $S_1 = 6$, what is the value of S_5?

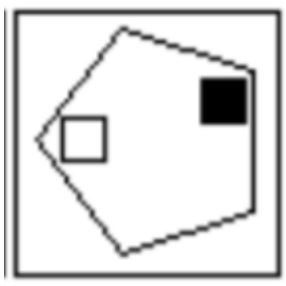
A. 16
B. 20
C. 24
D. 36

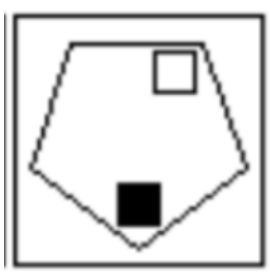
Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Spatial Aptitude

Which figure is identical to the first?

A.

B. 

C. 

D.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

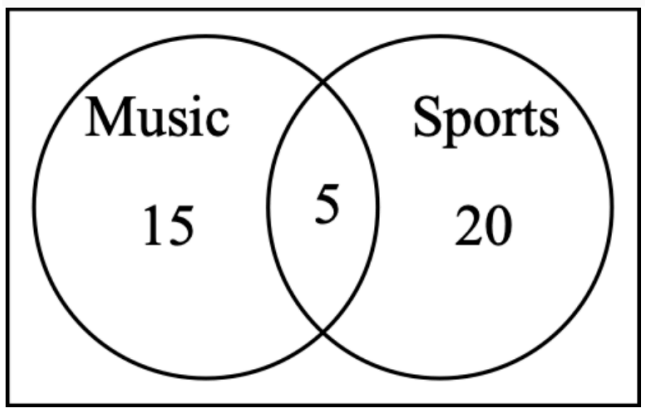
The ratio of zinc and copper in a brass pieces is $13:7$. How much zinc will be there in $100\ \mathrm{kg}$ of such a piece?

A. $20\ \mathrm{kg}$
B. $35\ \mathrm{kg}$
C. $55\ \mathrm{kg}$
D. $65\ \mathrm{kg}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

The results of asking 50 students if they participate in music or sports are shown in the Venn diagram.



What percentage of the 50 students do not participate in music and do not participate in sports?

A. 80%
B. 20%
C. 70%
D. 40%

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Rajiv Gandhi Khel Ratna Award was conferred _____ Mary Kom, a six-time world champion in boxing, recently in a ceremony _____ the Rashtrapati Bhawan (the President's official residence) in New Delhi.

A. with, at
B. on, in
C. on, at
D. to, at

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Bernard has wrapped gifts for the 14 students in his class, and numbered the gifts from 1 to 14.

Bernard knows the weight of each gift. After wrapping each gift, Bernard realized that he accidentally dropped his phone in one of the boxes. In order to avoid unwrapping many gifts, he plans to do the following:

1. Divide the 14 gifts into 2 piles, with the smallest 7 gift numbers in the first pile and the largest 7 gift numbers in the second pile.
2. Weigh the first pile. If it is heavier than expected, retain the first pile. Otherwise retain the second pile.
3. Divide the retained pile into 2 piles containing as close to the same number of gifts as possible. If there is an odd number of gifts, the first pile will have one fewer gift than the second pile. The smallest gift numbers are in the first pile.
4. Repeat Steps 2 and 3 until the retained pile has a single gift that contains the phone. Open that gift to retrieve the phone.

In which of the following situations would Bernard weigh the fewest number of piles?

A. The phone is in gift number 3.

B. The phone is in gift number 13.
C. The phone is in gift number 8.
D. The phone is in gift number 6.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Last year, a magazine charged a $\$50$ subscription fee. This year, the price will be increased by $\$10$. If the magazine could lose 4 subscribers this year and still collect the same revenue as it did last year, how many subscribers did the magazine have last year?

A. 21
B. 22
C. 23
D. 24

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 05sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Machine A, which produces 15 golf clubs per hour, fills a production lot in 6 hours.

Machine B fills the same production lot in 1.5 hours. How many golf clubs does machine B produce per hour?

A. 30
B. 40
C. 60
D. 90

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

If x satisfies the equation $4^{8^x} = 256$, then x is equal to _____.

A. $\frac{1}{2}$
B. $\log_{16} 8$
C. $\frac{2}{3}$
D. $\log_4 8$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

I am not sure if the bus that has been booked will be able to _____ all the students.

A. sit
B. deteriorate
C. fill

D. accommodate

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Graph Theory

Given a graph G with vertex set V and edge set E, which of the following is/are true if and only if G is a tree?

A. For all vertices u and $v \in V$, there exists a unique simple path from u to v.
B. G is connected, but if any edge $e \in E$ is removed, the resulting graph is not connected.
C. G is acyclic, but if any edge $e \notin E$ is added, the resulting graph is not acyclic.
D. $|E| = |V| - 1$ and either G is acyclic or G is connected.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Theory of Computation

Consider the following language L over the alphabet $\mathrm{A} = \{0, 1\}$. $L = \{w \in A^* : w$ does not contain any 1 's in even positions, where the leftmost is position $1\}$ The number of states in the minimal deterministic finite state automaton for L is ?? (Write answer 0 if L is not regular)

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Which of the following problems is/are decidable?

A. Determining whether two nondeterministic finite automata accept the same language.
B. Determining whether two context-free grammars represent exactly the same language.
C. Determining whether a propositional logic expression is satisfiable.
D. Determining whether a Turing machine will halt for a given input.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Consider a language L that is recognized by a machine M. Which of the following statements is/are not necessarily true?

A. If M is a deterministic finite automaton, then L can be represented by a regular expression.
B. If M is a non-deterministic finite automaton, then L can be represented by a context-free grammar.
C. If M is a deterministic pushdown automaton, then L can be represented by a context-free grammar.
D. If M is a Turing machine, then L is recursive.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Consider the following sequence of instructions intended for execution on a stack machine. Each arithmetic operation pops the second operand, then pops the first operand, operates on them, and then pushes the result back onto the stack.

```
    push b
    push x
    add
    pop c
 5. push c
    push y
    add
    push c
    sub
10. pop z
```

Which of the following statements is/are true?

I. If push and pop instructions each require 5 bytes of storage, and arithmetic operations each require 1 byte of storage, then the instruction sequence as a whole requires a total of 40 bytes of storage.
II. At the end of execution, z contains the same value as y.
III. At the end of execution, the stack is empty.

A. I, II, and III
B. II only
C. II and III only
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

A function returns a two's-complement two-byte integer. On success, the return value exceeds 1024. On error, the return value is in the range -1 through -10 to represent ten different error conditions.

Which of the following hexadecimals could not be returned by this function? (Select all that apply)

A. 0401
B. 03F0
C. FFED
D. FFF6

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Which of the following statements is/are true? ($\otimes$ is XOR operator, $*$ is AND operator)

I. $a \otimes b$ always equals $(\overline{a} * b) + (a * \overline{b})$
II. $a \otimes (b \otimes c)$ always equals $(a \otimes b) \otimes c$
III. $(a + \overline{b}) * (b + c)$ always equals $a + c$

A. III only
B. I and II only

C. II and III only
D. I, II, and III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Consider a memory of 32 blocks (labeled 0 through 31) and a cache of 8 blocks (labeled 0 through 7).

In the following sequence of memory block references from the CPU, beginning from an empty cache, on which reference must the cache layout for a direct-mapped and a 4-way set associative cache first differ? (i.e. the referenced memory block causes an existing cached block to be evicted from the cache)

order of reference	1	2	3	4	5	6	7	8
block referenced	0	15	18	5	1	13	15	26

A. Reference 5
B. Reference 2
C. Reference 4
D. Reference 6

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a machine with a physical memory of $8\ \mathrm{GB}$, a page size of $8\ \mathrm{KB}$, and a page table entry size of 4 bytes. How many levels of page tables would be required to map a 46-bit virtual address space if every page table fits into a single page?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a system with the following specifications:

- TLB has a 1 cycle hit time and a 99% hit rate
- The TLB miss penalty (page table lookup time) is 100 cycles
- The $\mathrm{L1}$ cache has a hit time of 4 cycles and a hit rate 80%
- The miss penalty for $\mathrm{L1}$ cache is 50 cycles

Assuming TLB and $\mathrm{L1}$ cache accesses do not overlap, what is the average memory access time for the system described above?

A. 15 cycles
B. 16 cycles
C. 12 cycles
D. 13 cycles

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

A binary single-precision floating point number contains the sequence of bits 10001111100000000001000000000000.

Information is stored in the following left-to-right order: sign bit, exponent (bias -127), and mantissa (with an implied unit bit).
Which of the following representations in decimal is equivalent?

A. $2^{31} * (1 + 2^{-12})$
B. $-1 * 2^{31} * (1 + 2^{-12})$
C. $-1 * 2^{-65} * (1 + 2^{-10})$
D. $-1 * 2^{-96} * (1 + 2^{-11})$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following two statements:

- **S1:** Consider an undirected, simple graph $G = (V, E)$ with edge weights $w_e \geq 0$ such that $w_e \neq w_{e'}$ for all $e, e' \in E, e \neq e'$. Let $s \in V$ be fixed. Then for all $v \in V$ there is a unique shortest (s, v)-path.

- **S2:** Consider an undirected, simple graph $G = (V, E)$ with edge weights $w_e \geq 0$ such that $w_e \neq w_{e'}$ for all $e, e' \in E, e \neq e'$. Then there is a unique minimum spanning tree T.

Which of the following is correct?

A. $\mathbf{S1}$ is true, but $\mathbf{S2}$ is false.
B. $\mathbf{S1}$ is false, but $\mathbf{S2}$ is true.
C. Both $\mathbf{S1}$ and $\mathbf{S2}$ are true.
D. Both $\mathbf{S1}$ and $\mathbf{S2}$ are false.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Given the following C code:

```
    int p=54,q = 4,r = 5;
    switch (A) {
        case 4: printf("%d", p);
        case 18: printf("%d", q);
5.      case 20: break;
        case 14: printf("%d", r);
        default: printf("%d", q);
    }
```

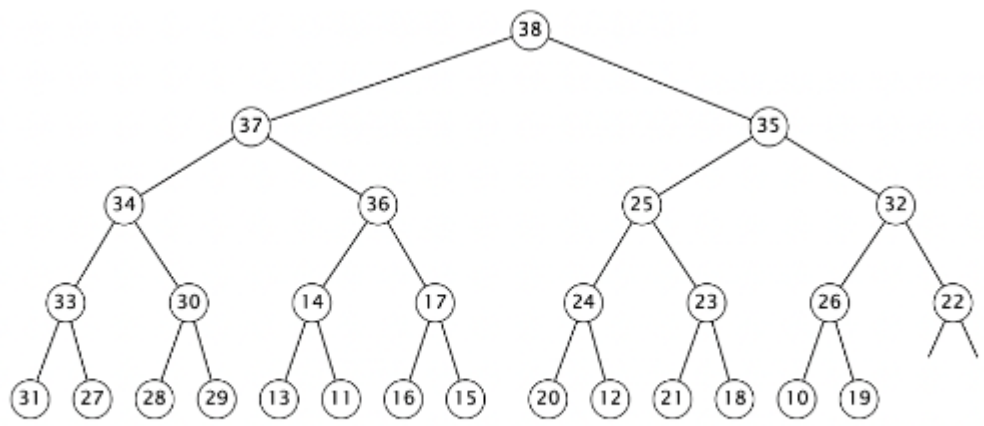
Choose the correct expression for $\mathbf{A}$ such that the output of the given $\mathbf{C}$ code fragment is 54.

A. $q * r$
B. $q * r - 2$
C. $q + r * 2$
D. $(q * r) - (q + 2)$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Consider the following binary heap.



Suppose that the last operation performed in the binary heap above was inserting the key x. What is/are possible values of x?

A. 19
B. 26
C. 32
D. 22

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider these 2 statements -

- $\mathrm{S1}$: If A is invertible, then A and A^{-1} have the same eigenvectors.
- $\mathrm{S2}$: If A is invertible, then A and A^{-1} have the same eigenvalues.

Which of the following is the correct option?

A. $\mathrm{S1}$ is correct but $\mathrm{S2}$ is incorrect
B. $\mathrm{S1}$ is incorrect but $\mathrm{S2}$ is correct
C. Both are correct
D. Both are incorrect

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose that X is a random variable where:

$$\mathrm{P}(\mathrm{X}=-3)=1/3 \quad \mathrm{P}(\mathrm{X}=0)=1/3 \quad \mathrm{P}(\mathrm{X}=3)=1/3.$$

and let $\mathrm{Y}=\mathrm{X}^2$.
What will be the value of $\mathrm{E}[\mathrm{Y}]$?

A. 6
B. 0
C. $2/3$
D. 1

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

To delete a node X with 2 non-null children in a BST, we replace the node X with the minimum node Y from X's right subtree. Which of the following is true about the node Y?

A. Y is always a leaf node
B. Y is always either a leaf node or a node with empty left child
C. Y may be an ancestor of the node
D. Y is always either a leaf node or a node with empty right child

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The intent of the function below is to delete the last node of the list.

```
void removeLast(Node* first) {
    Node *p, *q;
    p = first;
    q = p->next;
    while (q->next != NULL) {
        p = q;
        q = q->next;
    }
    p->next = NULL;
    free(q);
}
```

Which of the following describes the class of all linked lists for which this function works correctly?

A. No linked lists
B. All non-empty linked lists
C. All linked lists with more than one node
D. The empty list and all linked lists with more than one node

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Consider the following initialization of an array and pointer.

```
char a[] = "abc\0de";
char  *p = a ;
```

Let the system be 32 bit addressable. The size of int is 4 bytes and the size of char is 1 byte.
Which of the following is/are TRUE?

A. `sizeof(a) = 7`

B. `sizeof(p) = 4`

C. `sizeof(*p) = 1`

D. `strlen(p) = 3`

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Assume that an int variable takes 4 bytes and a char variable takes 1 byte. What is the output of the code below?

```
int main()
{
    int arr[]={10,20,30,40,50,60};
    int *ptr1=arr;
    int *ptr2=arr+5;
    printf("Number of elements between two pointer are: %d.",
    (ptr2 - ptr1));
    printf("Number of bytes between two pointer are: %d.",
    (char*)ptr2 - (char*)ptr1);
}
```

A. Number of elements between two pointer are: 5. Number of bytes between two pointers are: 20
B. Number of elements between two pointer are: 20. Number of bytes between two pointers are: 20
C. Number of elements between two pointer are: 5. Number of bytes between two pointers are: 5
D. Compile time error

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Consider the following grammar over the alphabet $\{a, b, d\}$:

$$S \rightarrow ABD$$
$$A \rightarrow a \mid BSB$$
$$B \rightarrow b \mid D$$
$$D \rightarrow d \mid \epsilon$$

What is the First set for the non-terminal S?

A. a, b, d
B. a, b
C. a, d
D. d, ϵ

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

An $\mathrm{LR}(1)$ parser:

1. A) Processes the input symbols from left to right.
2. B) Produces a left-most derivation of the input string.
3. C) Looks ahead at most one input symbol to decide the next parsing action.
4. D) Has a time complexity proportional to the number of input symbols, typically $O(n)$, where n is the length of the input string.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Which of the following statements about datagrams sent by a node in a network using IPv4 protocol is (are) true?

I. Datagrams at the source must be the size of the smallest maximum transmission unit (MTU) of all the links on a path to the destination.
II. Datagrams may be fragmented during routing.
III. Datagrams are reassembled only at the destination.

A. I only
B. II only
C. III only
D. II and III

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Consider a TCP connection utilizing cumulative acknowledgment and the sliding window protocol. Suppose the sender has most recently transmitted a segment with **ACK=500**, and then the receiver receives a segment with **SEQ=300** containing 50 bytes of data. How should the receiver respond in this scenario?

A. Send a segment with $ACK = 301$.
B. Send a segment with $\mathrm{ACK} = 350$.
C. Send a segment with $\mathrm{ACK} = 500$.
D. Send a segment with $ACK = 551$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Say that A is the primary key of table $\mathbf{R}$, and that B is another, numeric attribute of $\mathbf{R}$. Let r be the number of tuples in table $\mathbf{R}$.

$$(\pi_{\mathrm{A}}\,(\sigma_{\mathrm{B}<5}(\mathbf{R}))) \bowtie (\pi_{\mathrm{A}}\,(\sigma_{\mathrm{B}\geq 5}(\mathbf{R})))$$

How many tuples will result from the above RA expression?

A. 0
B. 1
C. $\frac{1}{2}r$
D. r

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Twelve page requests occur in the following order: $9, 36, 3, 13, 9, 36, 25, 9, 36, 3, 13,$ and 25. Assume that physical memory initially starts empty and fully-associative paging is used. Which of the following statements is/are true?

I. If the physical memory size is 3 pages, then most-recently-used (MRU) paging will result in the same number of faults as if the optimal algorithm was used.
II. If first-in-first-out (FIFO) paging is used, and the physical memory size is raised from 3 pages to 4 pages, then Belady's anomaly will appear.
III. If least-recently-used (LRU) paging is used, and the physical memory size is raised from 3 pages to 4 pages, then Belady's anomaly will appear.

A. II only
B. I and II only
C. II and III only
D. I, II, and III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

The routing table below uses the longest prefix match in its routing decisions.

Network Address/Length	Next Hop
196.94.8.0/21	S
196.94.16.0/24	T
196.94.0.0/16	U

What are the next hops for each of the following two destination addresses?

	Next Hop to 196.94.10.7	Next Hop to 196.94.34.9
(A)	S	T
(B)	S	U
(C)	T	T
(D)	T	U

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

We know that distance vector may converge slowly (i.e. count to infinity) if a link breaks. Can this problem also occur if a new link is added?

A. No, since adding the link can only lower the cost of best paths.
B. No, since the distance vector algorithm would simply not use this link.
C. Yes, just like in the case of link failures, outdated information may still bounce around in the network.
D. Yes, in fact the problem will become worse as there are more links on which counting-to-infinity can occur.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Suppose the 4000 -byte IP packet run into a link of MTU $= 1500$ bytes first and then a link of MTU $= 900$ bytes next. What is TRUE about fragments sent out on the MTU $= 900$ bytes link?

A. All Fragments has same ID number
B. There will be total 6 fragments

C. Offset of last fragment will be 3840
D. Total length of first, third and fifth fragments are same

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following lexical analyser "flex" specification -

	Regular Expression	Token
Rule 1	c	{ printf (" 1 "); }
Rule 2	ac^+b^*	{ printf (" 2 "); }
Rule 3	cc	{ printf (" 3 "); }
Rule 4	ab^*	{ printf (" 4 "); }
Rule 5	a	{ printf (" 5 "); }
Rule 6	$ac^*\ b^+$	{ printf (" 6 "); }

When using regular expressions to scan an input, we resolve conflicts by taking the largest possible match at any point.

If two longest matches are of the same length then we give preference in the below order: Rule 1(Highest Priority), Rule 6(Lowest Priority).

Which of the following rule(s) can be deleted such that lexer's behavior does not get changed for any of the input strings?

A. Rule 3
B. Rule 4
C. Rule 5
D. Rule 6

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following grammar where S is the start non-terminal and the productions for the non-terminals A, B, and C are not shown. There are no other productions for S:

$$S \rightarrow AB \mid AC$$
$$A \rightarrow \ldots$$
$$B \rightarrow \ldots$$
$$C \rightarrow \ldots$$

Assume the row corresponding to A, B, and C does not have any conflicting entries in the $\mathrm{LL}(1)$ parsing table.
What conditions must be satisfied by the First and Follow sets of the non-terminals such that the grammar is $\mathrm{LL}(1)$?

A. If $\mathrm{First}(A) \neq \{\epsilon\}$, grammar can never be $\mathrm{LL}(1)$.
B. If $\mathrm{First}(A) \neq \{\epsilon\}$, then grammar is $LL(1)$ by ensuring that $\mathrm{First}(B) \cap \mathrm{First}(C) = \emptyset$.
C. If $\mathrm{First}(A) = \{\epsilon\}$, $\epsilon \in \mathrm{First}(B)$ and $\epsilon \in \mathrm{First}(C)$, grammar can never be $\mathrm{LL}(1)$.
D. If $\mathrm{First}(A) = \{\epsilon\}$, and $\epsilon \in \mathrm{First}(B)$ and $\epsilon \in \mathrm{First}(C)$, then grammar is $LL(1)$ by ensuring that $\mathrm{First}(B) \cap \mathrm{First}(C) = \emptyset$.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Assume that any n-bit positive integer x is stored as a linked list of bits so that the first element of the list is the least significant bit. For example, $x = 14 = 1110_2$ is stored as the linked list $(0, 1, 1, 1)$ of size $n = 4$. For this data structure, the operation that replaces x by $\left\lfloor \frac{x}{8} \right\rfloor$ can be done in

A. $\Theta(1)$ steps
B. $\Theta(\log n)$ steps
C. $\Theta(n)$ steps
D. $\Theta(n \log n)$ steps

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following C program:

```c
#include <stdio.h>
int process(int n) {
    int x = 2;
    while (x < n) {
        if (n % x == 0) {
            n = n / x;
            x = 2;
            // Point A
        } else {
            x++;
            // Point B
        }
    }
    return n;
}
```

Consider two conditions: " $x < n$ " and " $x > 2$ "

Which of the following statements are TRUE regarding the conditions at Point $\mathbf{A}$ and Point $\mathbf{B}$?

A. At Point A, $x < n$ is SOMETIMES possible but not always.
B. At Point A, $x > 2$ is SOMETIMES possible but not always.
C. At Point B, $x < n$ is SOMETIMES possible but not always.
D. At Point B, $x > 2$ is SOMETIMES possible but not always.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Numerical Type | Award: 2 | Penalty: 0 | Programming in C

```c
#include <stdio.h>
int fun(int *i) {
    *i += 5;
    return 2;
}
int main() {
    int x = 3;
    if ((x == 2) && (fun(&x))) {
        x++;
    }
    x = x + fun(&x);
    printf("%d\n", x);
    return 0;
}
```

What will be the output for the above program?

Assume Left-to-right operand evaluation for "+" operator.

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

You have n lists, each consisting of m integers sorted in ascending order. Merging these lists into a single sorted list will take time:

A. $O(nm \log m)$
B. $O(mn \log n)$
C. $O(m+n)$
D. $O(mn)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

In a network graph $G = (V, E)$, each edge $e \in E$ has a probability p_e of successful transmission. Given two vertices $s \neq t \in V$, your goal is to find the s - t path with the **highest probability of successful transmission.**

To achieve this, you construct a new graph $G' = (V', E')$ with $V' = V$ and $E' = E$, and assign weights to edges in G' so that finding the shortest path in G' **using Dijkstra's algorithm** will correspond to finding the **highest probability path in** G.

Which of the following could be appropriate weights for edge e in G' if its probability in G is p_e? (Select all that apply.)

A. p_e
B. $\log(p_e)$
C. $\log\left(\frac{1}{p_e}\right)$ or $-\log(p_e)$
D. p_e^2

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following function.

```
Function EXPL( int n ) {
    if ( n <= 2 ) then return 1
    return EXPL( n / 2 ) * lg( n )
}
```

What does EXPL (n) return, if n is a power of 2?

A. $\lg(n)$
B. $n * \lg(n)$
C. $n! * \lg(n)$
D. $(\lg(n))!$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose Z_1, Z_2, Z_3 are i.i.d. (independently and identically distributed) Bernoulli random variables with parameter $p = \frac{1}{2}$. The random variable $S = Z_2 Z_3$ if $Z_1 = 1$, and $S = Z_2 + Z_3$ if $Z_1 = 0$. Find $P(Z_1 = 1 \mid S = 1)$.

A. $1/8$
B. $1/3$
C. $3/8$
D. $1/2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $f(3) = 3$ and $f'(x) \geq 2$ for $3 \leq x \leq 6$, use the Mean Value Theorem to determine the smallest possible value of $f(6)$.

A. 2
B. 3
C. 6
D. 9

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Consider a binary function $g : M \times M \rightarrow \{$ true, false $\}$, where M is a non-empty subset of the natural numbers that contains an even number of distinct elements.

Which of the following statements can never be true about g? (Select all that apply)

A. g is symmetric and antisymmetric.
B. g defines a total order and an equivalence relation.
C. g is an injection
D. g is antisymmetric but not a surjection

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

A certain function $g : N \times N \rightarrow N$ is defined as follows for x and $y \geq 0$:

$$\begin{cases} g(0,y) = y+1 & \text{for } y \geq 0 \\ g(x,0) = g(x-1,1) & \text{for } x > 0 \\ g(x,y) = g(x-1, g(x,y-1)) & \text{for } x \text{ and } y > 0 \end{cases}$$

Which of the following statements is/are true?

I. $g(1,n) = n-3$
II. $g(2,n) = 2n-3$
III. $g(3,n) = 2^{n+3}-3$

A. III only
B. II and III only
C. I, II, and III
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

For a pair (A,B) of subsets of the set $X = \{1,2,\ldots,100\}$, let $A \triangle B$ denote the set of all elements of X which belong to exactly one of A or B. The number of pairs (A,B) of subsets of X such that $A \triangle B = \{2,4,6,\ldots,100\}$ is

A. 2^{151}
B. 2^{102}
C. 2^{150}
D. 2^{100}.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Given a predicate P, and a domain D, we define the **truth set** of P to be the set of elements x in D for which $P(x)$ is true. The truth set of $P(x)$ is denoted by $\{x \in D \mid P(x)\}$.

Similarly, the **falsity set** of a predicate is the set of elements in D for which the predicate is false.

Let P and Q be two predicates, and let their truth sets be denoted as T_P, T_Q, and falsity sets be denoted as F_P, F_Q.

Which of the following identities are correct?

A. $T_P \cap T_Q = T_{P \wedge Q}$
B. $T_P \cup T_Q = T_{P \vee Q}$
C. $F_P \cap F_Q = F_{P \vee Q}$
D. $F_P \cup F_Q = F_{P \wedge Q}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Consider the following grammars:

Grammar G1:
$S \rightarrow 0\ T \mid \varepsilon$
$T \rightarrow 1\ S$

Grammar G2:
$S \rightarrow T\ S$
$S \rightarrow \varepsilon$
$T \rightarrow X\ Y$
$X \rightarrow 0$
$Y \rightarrow 1$

Which of the following statements is/are not true?

A. Grammar $G1$ can generate any string that $G2$ can, & Grammar $G2$ can generate any string that $G1$ can.
B. Grammar $G1$ is ambiguous.
C. Grammar $G2$ is ambiguous.
D. A regular grammar is never ambiguous.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Suppose that process P has been running for several days when a new process Q starts up and begins contending with P for resources. Which of the following is/are true?

A. In a wait-die system, if P needs a resource held by Q, then P waits.
B. In a wait-die system, if Q needs a resource held by P, then Q waits.
C. In a wound-wait system, if P needs a resource held by Q, then Q yields and waits.
D. In a wound-wait system, if Q needs a resource held by P, then P yields and waits.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

A relation $R(A, B, C, D)$ is decomposed into ACD and BC. Which of the following statements is/are true? (Consider each scenario independently.)

A. The decomposition is lossless if C is a candidate key for ACD.
B. If the decomposition is lossy, the natural join of ACD and BC will be missing tuples found in R.
C. If the decomposition is lossless, then both ACD and BC could have strictly fewer rows than R.
D. If the decomposition is lossless, then both ACD and BC have the same number of rows as R.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A certain hard drive rotates at 6000 rpm . It has 1 KB per sector and averages 128 sectors per track. Which of the following statements is/are true?

I. The average latency of the drive is under 6 milliseconds.
II. The burst data rate of the drive is over $10\mathrm{MB}/\mathrm{sec}$.
III. The average capacity per track is over 1 MB.

A. II only
B. I and II only
C. II and III only
D. I, II, and III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A super-simple RISC-like computer has three main computational elements (memory, registers, and ALU) which it uses to implement the four types of instructions shown below. This chart shows which main computational elements each instruction uses. For example, arithmetic instructions first require accessing memory, then accessing registers, then accessing the ALU, then accessing registers once more.

	Instruction Memory (Instruction Fetch)	Registers (Decode + Register Read)	ALU (Execute)	Data Memory (Data Access)	Registers (Write Back)
Branch	■	■	■		■
Arithmetic	■	■	■		■
Load	■	■	■	■	■
Store	■	■	■	■	

Suppose these computational elements require the following time to operate and settle (for either read or write operations):

- Memory: M ns
- Registers: R ns
- ALU: A ns

Which of the following statements is/are true?

I. If the machine has a single-cycle datapath with a constant cycle time, then the cycle cannot be faster than $2 * M + 2 * R + A$ ns.
II. If the machine has a multi-cycle datapath with a constant cycle time, then operations can be pipelined, but the cycle cannot be faster than $\max(\mathrm{M}, \mathrm{R}, \mathrm{A})$ ns.
III. Suppose that the machine has a single-cycle datapath, but the cycle length is varied as follows: Branch and arithmetic instructions will execute in $M + 2 * R + A$ ns; load instructions will execute in $2 * M + 2 * R + A$ ns ; store instructions will execute in $2 * M + R + A$ ns. In this case, the throughput will be better than in option I or II above.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

While solving the system of linear equations, the three elementary row operations on a matrix are as following :

1. Row Swap: A row within the matrix can be switched with another row.
2. Row Scaling: Each element in a row can be multiplied by a non-zero constant.
3. Row Replacement: A row can be replaced by the sum of that row and a multiple of another row.

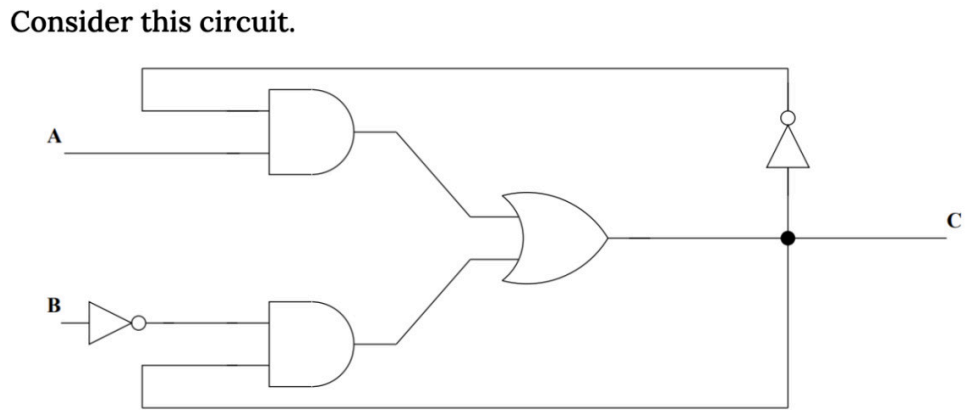
Which of the following is true regarding the Elementary Row Operations?

A. Row Swap can be performed by using the other two elementary row operations.
B. Row Scaling can be performed by using the other two elementary row operations.
C. Row Replacement can be performed by using the other two elementary row operations.
D. None of the elementary row operations can be performed using the other two.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Consider this circuit.



Suppose that the circuit is initialized by setting input $\mathrm{A} = 0$ and $\mathrm{B} = 0$, with $\mathrm{C} = 0$. Assume that all inputs and outputs are clocked. Which of the following statements is/are true?

I. After the circuit is initialized, if A is set to 0 and B is set to 1, then output C becomes 1 in the next clock pulse.
II. After the circuit is initialized, if A is set to 1 and B is set to 0, then output C becomes 1 in the next clock pulse.
III. After the circuit is initialized, if A is set to 1 and B is set to 1, then output C becomes 1 in the next clock pulse.

A. II only
B. III only
C. II and III only
D. I, II, and III

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

When we list 2^n distinct n-bit binary numbers in a Gray code, adjacent numbers differ by exactly one bit.

In principle, there can be more than one such code for a given word length.

Consider the following Gray code sequence.

$$000, 100, 101, 001, 011, \mathrm{X}, 110, \mathrm{Y}$$

What are the missing entries, X and Y?

A. $X = 010, Y = 111$
B. $X = 010, Y = 100$
C. $X = 111, Y = 010$
D. $X = 111, Y = 100$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following functional dependencies in a database.

$$\mathrm{AB} \rightarrow \mathrm{DE}, \mathrm{C} \rightarrow \mathrm{E}, \mathrm{D} \rightarrow \mathrm{C}, \text{and } \mathrm{E} \rightarrow \mathrm{A}$$

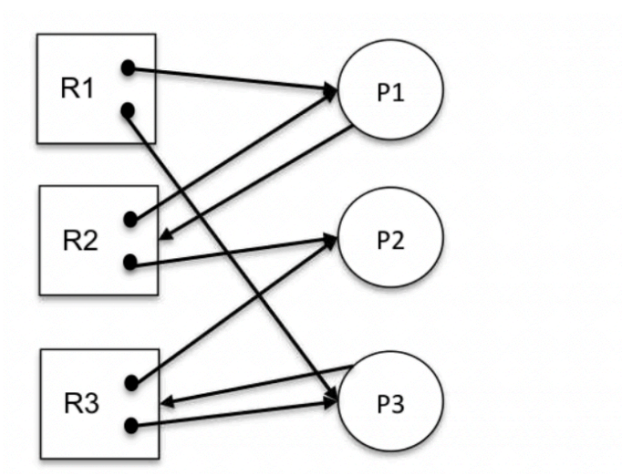
The relation $\mathrm{R}(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{E})$ is

A. in second normal form but not in third normal form
B. in third normal form but not in BCNF
C. in BCNF
D. not in second normal form

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following resource allocation graph:



Which of the following is/are correct about the above resource allocation graph?

A. The current System is in a deadlock.
B. The current System is in deadlock but adding one more instance of $\mathrm{R1}$ will make the system deadlock-free.
C. The current System is NOT in a deadlock but if we add an additional process $\mathrm{P4}$ that demands an instance of $\mathrm{R1}$, then the system will go to a deadlock.
D. The current System is NOT in a deadlock but if $\mathrm{P2}$ additionally demands one instance of $\mathrm{R1}$, then the system will go to a deadlock.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Suppose two threads execute the following C code concurrently, accessing shared variables a, b, and c :

Initialization

```
    int a = 4;
    int b = 0;
    int c = 0;
```

Thread 1

```
    if (a < 0) {
        c = b - a;
    } else {
        c = b + a;
5. }
```

Thread 2

```
    b = 10;
    a = -3;
```

You can assume that reads and writes of the variables are atomic, and that the order of statements within each thread is preserved in the code generated by the C compiler.

Statement $\mathrm{X} = \mathrm{Y} * \mathrm{Z}$ reads Y, then reads Z from memory, performs $\mathrm{Y} * \mathrm{Z}$, stores/writes it to memory X.

Which of the following values are possible for **c** after both threads complete?

A. 4
B. 7
C. 14
D. -3

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages over the alphabet $\mathrm{A} = \{0, 1\}$ is/are not regular?

A. $\{w \in A^* : w$ contains equal numbers of $1's$ and $0's\}$
B. $\{w \in A^* : w$ contains a prime number of $1's\}$
C. $\{w \in A^* : \exists u \in A^*$ such that $w = uu\}$
D. $\{\mathrm{w} \in \mathrm{A}^* : \exists \mathrm{u} \in \mathrm{A}^*$ such that $\mathrm{u} = \mathrm{ww}\}$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

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